Unlimited OCR Works: One-shot Long-horizon Parsing

Baidu Vision & Document Intelligence Report (arXiv:2606.23050)

1. Core Motivation & Human Working Memory

End-to-end OCR models using an LLM decoder suffer from linear KV cache growth as output length increases. When parsing 20 to 40 pages of PDF documents, standard Multi-Head Attention (MHA) accumulates tens of thousands of text tokens, causing severe GPU memory pressure and latency spikes.

Unlimited-OCR replaces all decoder attention layers with Reference Sliding Window Attention (R-SWA). Every generated token attends globally to all visual reference tokens (m = 256 per 1024x1024 page), while restricting causal text attention to a sliding window of n = 128 tokens.

2. Mathematical Formulation of R-SWA

```
KV_Cache(R-SWA) = L_prefix (visual + prompt) + n (sliding window = 128)
Cache_Ratio(T)  = (L_m + n) / (L_m + T)  -->  Constant O(1) decode footprint
as T >> n
```

3. Architecture Specifications (DeepSeek-OCR MoE Baseline)

Module / Component	Configuration	Engineering Benefit
Vision Tower (DeepEncoder)	SAM-ViT + CLIP-ViT (16x compression)	
Language Decoder (MoE)	3.3B Total / 500M Active per token	
Decoder Attention (R-SWA)	Global Prefix + 128-token Window	Flat FlashAttention-3 latency curve
Max Context Window	32,768 tokens (Base & Gundam modes)	

4. OmniDocBench v1.5 / v1.6 Benchmark Comparison

Restricting the text sliding window to 128 tokens does not degrade accuracy. Instead, avoiding long-range text attention divergence improves table structure recognition (TEDS +5.96%) and reading order accuracy.

Model Name	Active Params	OmniDocBench v1.5	OmniDocBench v1.6	
DeepSeek-OCR (Baseline)	500M (MoE)	87.00%	88.45%	4,951 tokens/s
DeepSeek-OCR 2	500M (MoE)	90.85%	91.60%	5,120 tokens/s
baidu/Unlimited-OCR	500M (MoE)	93.00% (+6.0%)	93.92% (SOTA)	

5. Long-Horizon Decoding Speed Stability (TPS vs Output Length)

At 256 generated tokens, Unlimited-OCR and DeepSeek-OCR run at nearly identical speeds. However, by 6,000 generated tokens, DeepSeek-OCR suffers a 35% throughput drop due to linear KV cache accumulation, while Unlimited-OCR maintains a constant generation speed across 20 to 40+ pages.

6. Resolution Modes & Practical Engineering Caveats

1) Single-page Gundam Mode (base_size=1024, image_size=640, crop_mode=True):
Uses dynamic high-resolution cropping for dense single-page tables, formulas, and tiny footnotes.

2) Multi-page Base Mode (image_size=1024, crop_mode=False):
Used for one-shot PDF parsing. Pages are rendered at 300 DPI via PyMuPDF and concatenated with <page>.

3) N-gram Anti-Repetition Guard (no_repeat_ngram_size=35):
Must be enabled (window=128 for single page, 1024 for multi-page) to prevent loops on blank borders.

7. Quick Local Deployment Commands

```
# Option A: vLLM OpenAI-compatible Server
vllm serve "baidu/Unlimited-OCR"

# Option B: Apple Silicon Mac M1 Pro (MLX 4-bit / 8-bit)
python -m mlx_vlm generate --model sahilchachra/unlimited-ocr-mxfp4-mlx --image
page_0001.png
```